Arithmetic Expression Geometry III: Singular Zero Geometry and Tubes

Multi-Zero Constructions, Discriminants, and Topological Transport

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Abstract

Arithmetic expression geometry associates a real assignment to an oriented Riemannian surface and imposes the eikonal identity $|\nabla a|_g^2 = \mu^2 + \lambda^2 a^2$, with $\mu \neq 0$. Its regular zeros are curves, not isolated roots. This paper develops the singular and parameterized zero geometry needed to turn those curves into genuine tubes, while separating them from the finite-root braids carried by complex-valued fields.

We first prove a conformal realization theorem: every smooth submersion on a surface becomes a regular arithmetic expression space after an explicit metric rescaling. This yields verified models with any prescribed finite number of parallel zero components and a logarithmic-cover model with countably many lifted sheets. A rank-r parameterized zero-section theorem then gives the exact transversality and properness hypotheses under which a zero incidence is a smooth fiber bundle. For real assignments over a circle with globally oriented vertical tangent bundle, every compact boundaryless proper zero tube is a union of tori. An explicit helical family computes its component permutation, mapping-torus decomposition, logarithmic deck shifts, and torus-link traces.

Topology change is confined to a discriminant. Conformal Morse families realize birth/death and reconnection in the singular category; the complex model $w^2 = \tau$ instead describes a branched filling whose boundary is a regular half-twist. Using the arithmetic-holomorphic coordinate of Paper II, every braid is realized by a square-free polynomial root family. Finally, affine conjugation recovers an Alexander quandle. Its ordinary torsion cocycle is exact, whereas a finite-field resonant twisted version represents a nonzero twisted cohomology class. Nevertheless, we prove that its standard state sum for every planar classical link collapses to the Alexander-quandle coloring count. Thus it is not evidence of a new invariant beyond Alexander or Burau. Thread selection, Markov-normalized AEG decorations, and any stronger separation theorem remain open.

Keywords: arithmetic expression geometry; zero set; singularity; fiber bundle; braid group; logarithmic covering; twisted quandle cohomology; knot invariant

Contents

1 Singular AEG and the hierarchy of zero objects

1.1 The question

Paper I associates to continuous affine evaluation a regular arithmetic expression space

$$(M, g, a; \mu, \lambda), \qquad |\nabla a|_g^2 = \mu^2 + \lambda^2 a^2, \qquad \mu \neq 0, \tag{AEG}$$

where M is an oriented smooth surface, g is Riemannian, and a is a real-valued assignment function [9]. The right-hand side of (AEG) is strictly positive. Thus a is a submersion, and every nonempty level set—in particular $a^{-1}(0)$—is a regular curve. An isolated zero, a crossing, a branch point, or a birth/death event cannot be a regular interior zero.

The word *tube* is nevertheless natural. If an assignment varies with a parameter, its zero curves sweep a surface in the product of the arithmetic surface and the parameter space. The central problem is to determine when that swept surface is more than a picture: when it is smooth, when its projection is a fiber bundle, where topology may change, and what additional data are needed before a braid or knot can be extracted.

The historical AEG notes used "tube" for several disjoint unions of ambient geometries. A source audit of the repository and its Git history found no verifiable general construction under the old symbols E_k or $E_{\log}$. Accordingly, this paper does not import those symbols as established objects. It constructs descriptive finite-zero, logarithmic-cover, and helical models from first principles. The distinction matters: an ambient family is not its zero incidence, and a smooth zero incidence is not automatically a tube.

1.2 Imported interfaces and new scope

We use three results of Papers I and II as interfaces rather than redefining them.

1. Paper I proves the regular-zero theorem just recalled and defines a *singular AES*. Its singular set is closed, nowhere dense, and locally essential, and the regular equation holds on the complement.

2. Paper I proves that spatial regularity along the total zero set of a smooth real family makes that total zero set a smooth submanifold and its parameter projection a submersion. It explicitly leaves global triviality open without properness.

3. Paper II equips every oriented regular AES with its underlying complex structure. On the complete basic hyperbolic model, the arithmetic-holomorphic coordinate $w = X + iy$ is a global complex coordinate, and arithmetic holomorphicity is ordinary holomorphicity in w [10].

The first two interfaces concern a real scalar assignment and hence codimension-one zero curves. The third permits complex-valued auxiliary fields and hence codimension-two isolated roots. They must not be conflated.

Definition 1.1 (Stratified singular AES)**.** A *stratified singular AES* is a singular AES in the sense of Paper I together with a specified locally finite Whitney stratification $\mathcal{S}$ of its singular set S. A stratum is called a *singular zero stratum* if it is contained in the closure of $a^{-1}(0) \cap (M \setminus S)$. The stratification is additional control data; it is not claimed to be canonical.

This refinement supplies a language for parameterized Morse examples and branched fillings. It does not solve the classification of singular AEG germs. Indeed, the conformal realization theorem in Section 2 shows that the bare singular-AES category is deliberately flexible.

1.3 Seven levels that must remain distinct

Let $q : X \to B$ be an ambient family and let s be a real or vector-valued field on X. The following hierarchy fixes our terminology.

Definition 1.2 (Zero-object hierarchy)**.** Write $\mathcal{Z}(s) = s^{-1}(0)$ and $\pi = q|_{\mathcal{Z}(s)}$.

1. $\mathcal{Z}(s)$ is the *total zero set*; no regularity is implied.
2. It is a *smooth zero incidence* if it is a smooth submanifold of X.
3. It is a *proper zero tube* if π is a surjective, proper smooth submersion (with the corresponding neat boundary hypotheses when boundaries are present).
4. It is an *embedded zero tube* when its inclusion in the specified ambient family is retained as part of the data.
5. A *threaded zero tube* is an embedded zero tube together with a specified finite multisection or other explicitly declared one-dimensional thread datum.
6. A *braid closure* is obtained only after a finite moving-point system, an ambient surface, and a closure convention have been fixed.
7. A *knot invariant* is a quantity on closures that has been proved independent of all presentation choices, in particular the appropriate conjugation and stabilization moves.

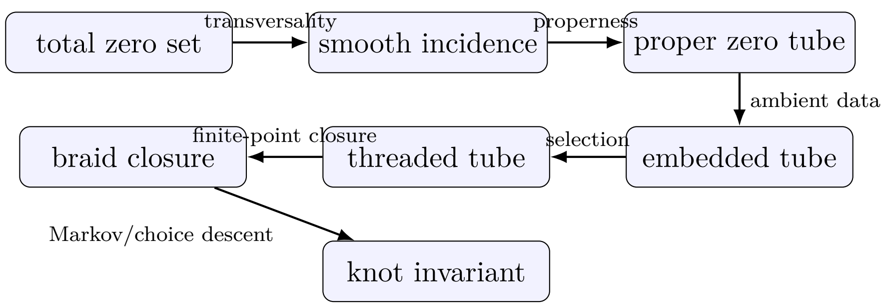

Figure 1: Each arrow requires new hypotheses or data. No arrow in the diagram is automatic from the object to its left.

The hierarchy also exposes a dimension obstruction. A real field on a surface has one-dimensional zero fibers; a complex field has zero-dimensional root fibers. Only the latter produces an ordinary geometric braid without an extra transverse probe.

1.4 Principal results

The paper proves the following claims. Status qualifications are included to make the mathematical boundary visible.

The positive and negative results are complementary. The realization theorems show that tube and braid geometry genuinely occur inside AEG. The flexibility and collapse theorems show that occurrence alone carries little discriminatory power. Any new invariant must come from a natural decoration tied to expression history, and that decoration must survive the relevant quotient.

Result	Content	Status
Conformal realization	Every smooth submersion on a surface satisfies (AEG) for an explicit conformal metric.	Proved
Multi-zero models	Explicit regular AESs have exactly k parallel zero components, or countably many logarithmic lifts.	Proved
Incidence theorem	Vertical transversality gives a smooth incidence and a submersion; properness gives a bundle.	Proved with stated hypotheses
Real-tube structure	Compact boundaryless real zero tubes over S^1 with global vertical orientation are unions of tori.	Proved with stated hypotheses
Helical tube	A compact family computes component permutation, logarithmic deck shift, and torus-link boundary traces.	Proved
Singular models	Definite and indefinite Morse events satisfy the AEG equation off an essential singular point.	Proved examples
Root monodromy	Square-free arithmetic-holomorphic polynomial families give proper root coverings and realize every braid.	Proved via Paper II
Affine novelty filter	Stateless scalars collapse to writhe; ordinary affine torsion is exact; resonant finite-field twisted torsion is nonzero but its planar state sum collapses to coloring count.	Proved
Beyond baselines	A Markov-descended, choice-free AEG decoration separates closures not separated by Alexander/Burau baselines.	Open

Table 1: Claim ledger for the main results.

1.5 Conventions

All manifolds are smooth, Hausdorff, and second countable. Unless boundary hypotheses are stated, manifolds are boundaryless. The parameters $\mu \neq 0$ and $\lambda \in \mathbb{R}$ always belong to the AEG equation; family parameters are denoted by b, t, θ, or τ. The letter t is also used for an affine multiplier only in Section 8, where it cannot be confused with a family coordinate. A deck transformation is denoted D, while T_{coef} denotes the Laurent-module action in twisted quandle cohomology.

For a loop of configurations, a based loop determines a braid element and a free loop determines a conjugacy class. "Invariant" without qualification means an ambient-isotopy invariant of oriented links; annular, transverse, framed, and presentation-dependent data are named explicitly.

2 Local zero models and conformal realization

2.1 The regular germ

At a regular zero p, the submersion theorem gives coordinates (u, v) centered at p in which $a = \mu u$: take $u = a/\mu$ and choose the sign of v so that (u, v) is positively oriented. This statement concerns the zero germ and the assignment; it does not put the metric into Euclidean form. Equation (AEG) further gives $|\nabla a|_g = |\mu|$ along $a = 0$. Thus a regular zero has one transverse direction with a fixed nonzero speed and one tangent direction. No finite-order singularity can occur there.

The simplest global realization keeps the metric fixed.

Example 2.1 (Cylindrical propagation)**.** On $M = \mathbb{R}_r \times S^1_\phi$ with $g = \mathrm{d}r^2 + \mathrm{d}\phi^2$, set

$$a_c(r,\phi) = \begin{cases} \dfrac{\mu}{\lambda}\sinh(\lambda(r-c)), & \lambda \neq 0, \\ \mu(r-c), & \lambda = 0. \end{cases}$$

Then $|\nabla a_c|^2_g = \mu^2 + \lambda^2 a_c^2$ and $Z(a_c) = \{c\} \times S^1$. This model will yield a proper product tube in Section 5.

2.2 Conformal realization

The following construction is elementary, but it determines the scope of every subsequent singularity claim.

Theorem 2.2 (Conformal realization)**.** *Let (M,h) be an oriented Riemannian surface, let $a \in C^\infty(M,\mathbb{R})$ be a submersion, and fix $\mu \neq 0$ and $\lambda \in \mathbb{R}$. The conformal metric*

$$g_a = \frac{|\,\mathrm{d}a|^2_h}{\mu^2 + \lambda^2 a^2}\, h \tag{1}$$

is smooth and positive definite, and $(M, g_a, a; \mu, \lambda)$ is a regular AES.

Proof. Put $g_a = \rho h$. Since $\mathrm{d}a$ never vanishes and the denominator is positive, ρ is a positive smooth function. The inverse metric is $g_a^{-1} = \rho^{-1}h^{-1}$, so

$$|\nabla a|^2_{g_a} = |\,\mathrm{d}a|^2_{g_a^{-1}} = \rho^{-1}|\,\mathrm{d}a|^2_h = \mu^2 + \lambda^2 a^2.$$

□

Corollary 2.3 (Singular realization)**.** *Let $a \in C^\infty(M,\mathbb{R})$ have a closed nowhere-dense critical set $S = \mathrm{Crit}(a)$. On $M \setminus S$, define g_a by (1). Then $(M, S, g_a, a; \mu, \lambda)$ is a singular AES. Every point of S is locally essential.*

Proof. Theorem 2.2 applies on $M \setminus S$. If a point $p \in S$ admitted a regular local extension agreeing with a, then $\mathrm{d}a_p = 0$ would imply $|\nabla a|^2(p) = 0$ for the extended nondegenerate metric, whereas the right-hand side of (AEG) is at least $\mu^2 > 0$. This is impossible. □

Warning 2.4 (Flexibility boundary). When the metric may be chosen as part of the construction, the regular AEG equation admits every smooth submersion, and the singular category admits every smooth function with closed nowhere-dense critical set. Therefore the bare equation does not by itself select an AEG-specific list of singularity types. A classification theorem would require additional naturality or analytic hypotheses, such as a fixed ambient metric, completeness, curvature bounds, controlled metric asymptotics, homogeneity, or derivation from expression history.

The warning is not a defect of the construction. It is a precise division of labor. The equation controls assignment speed relative to a chosen metric; it does not select that metric. Later tube theorems are consequently topological theorems with explicit hypotheses, not rigidity theorems for arbitrary AEGs.

2.3 Singular zero mechanisms

There are at least four mechanisms by which the regular-zero conclusion can fail.

1. The assignment may fail to be differentiable at a point, even though the metric extends there.
2. The assignment may be smooth but critical, forcing the conformal realization metric to degenerate or be removed at that point.
3. The domain may have a boundary or a puncture through which a zero component enters, exits, or escapes.
4. In a parameterized family, the total incidence may remain smooth while its projection to parameter space ceases to be a submersion.

Paper I supplies a verified example of the first mechanism. On the unit disc with $\rho = (\xi^2+\eta^2)^{1/2}$, it takes

$$g_{\mathbb{D}} = \frac{4}{\lambda^2(1-\rho^2)^2}(\,\mathrm{d}\xi^2 + \mathrm{d}\eta^2), \qquad a_{\mathbb{D}} = \frac{2\mu}{\lambda}\frac{\rho}{1-\rho^2} \quad (\rho > 0), \tag{2}$$

with $a_{\mathbb{D}}(0) = 0$ and $\mu, \lambda > 0$. The metric extends smoothly, the assignment is not C^1 at the center, and the center is its only zero. This is an isolated singular zero, not a normal form for all isolated singular zeros.

The second and fourth mechanisms will appear together in the Morse families of Section 6. The third is the reason properness and boundary control occur explicitly in Definition 1.2.

2.4 What local equivalence should preserve

For clarity, we distinguish three notions that are sometimes compressed into the phrase "same singularity."

Definition 2.5 (Three local equivalences)**.** For two pointed singular AEGs, a *zero-germ equivalence* is a local diffeomorphism carrying one zero germ to the other. An *assignment-germ equivalence* additionally intertwines the assignments up to a local diffeomorphism of the target fixing zero. An *AEG-germ isomorphism* preserves the assignment, the oriented metric, and the constants (μ, λ).

The real Morse lemma classifies assignment germs up to a much coarser right-left equivalence. Conformal realization then equips those germs with examples of AEG metrics. It does not show that all AEG-germ isomorphism classes are represented by the displayed metrics, nor that the metric asymptotics are unique.

Open problem 2.6 (Rigidity beyond conformal realization)*.* Find natural hypotheses derived from arithmetic evaluation under which the local singular AEG germ has a finite classification, and determine which of the Morse and branch models of Section 6 survive that restriction.

3 Multi-zero and logarithmic constructions

3.1 A finite and locally finite construction

The regular-zero theorem is local and does not limit the number of connected components globally. The next model gives a direct answer to the existence part of the multi-zero problem.

Theorem 3.1 (Parallel multi-zero model)**.** *Let $h \in C^\infty(\mathbb{R})$ have only simple zeros. On $M = \mathbb{R}^2$ define*

$$a_h(x, y) = e^x h(y) \tag{3}$$

and

$$g_h = \frac{e^{2x}(h(y)^2 + h'(y)^2)}{\mu^2 + \lambda^2 e^{2x} h(y)^2}(\,\mathrm{d}x^2 + \,\mathrm{d}y^2). \tag{4}$$

Then $(M, g_h, a_h; \mu, \lambda)$ is a regular AES and

$$Z(a_h) = \coprod_{h(c)=0} \mathbb{R} \times \{c\}. \tag{5}$$

In particular, if $h_k(y) = \prod_{j=1}^{k}(y-c_j)$ with distinct c_j, the zero set has exactly k connected components.

Proof. One has

$$\mathrm{d}a_h = e^x h(y)\ \mathrm{d}x + e^x h'(y)\ \mathrm{d}y, \qquad |\,\mathrm{d}a_h|^2_{\mathrm{eucl}} = e^{2x}(h^2 + (h')^2).$$

The simple-zero hypothesis implies that h and h' never vanish simultaneously. The coefficient in (4) is therefore smooth and positive. Theorem 2.2 proves the AEG identity, and e^x never vanishes, so (5) is immediate. □

Remark 3.2 (What the index counts)*.* The polynomial example proves existence of a regular AES with k zero components. It does not prove uniqueness, completeness, homogeneity, or a canonical classification by k. We therefore call it the *parallel k-zero model*, not E_k. The old repository symbol E_k had no stable definition: at different times its subscript suggested construction order, zero count, or singularity count.

Taking $h(y) = \sin y$ gives a countable, locally finite zero family. This example already suggests that countably many visible sheets may be repetitions under a covering group rather than independent components of downstairs geometry.

3.2 A downstairs model and its logarithmic pullback

Let $z = X + iY$ be the standard coordinate on $\mathbb{C}^\times$ and set

$$a_\times(z) = Y, \qquad g_\times = \frac{\mathrm{d}X^2 + \,\mathrm{d}Y^2}{\mu^2 + \lambda^2 Y^2}. \tag{6}$$

Since $|\,\mathrm{d}Y|^2_{\mathrm{eucl}} = 1$, this is a regular AES. Its zero set is the punctured real axis, hence has two components.

Pull (6) back along the exponential covering

$$\exp : \mathbb{C} \longrightarrow \mathbb{C}^\times, \qquad x + iy \longmapsto e^{x+iy}.$$

The resulting data are

$$\widetilde{a}(x, y) = e^x \sin y, \qquad \widetilde{g} = \frac{e^{2x}}{\mu^2 + \lambda^2 e^{2x} \sin^2 y}(\,\mathrm{d}x^2 + \,\mathrm{d}y^2), \tag{7}$$

and

$$Z(\widetilde{a}) = \coprod_{k\in\mathbb{Z}} \{y = k\pi\}. \tag{8}$$

Theorem 3.3 (Logarithmic zero lift). *The data in (7) form a regular AES and are invariant under the deck transformation*

$$D(x, y) = (x, y + 2\pi).$$

On the labels in (8), D acts by $k \mapsto k+2$. Thus the countable upstairs zero family has exactly two deck orbits, corresponding to the positive and negative real half-axes downstairs.

Proof. The formula is the pullback of the regular AES (6); it can also be checked directly from $|\,\mathrm{d}\tilde{a}|^2_{\mathrm{eucl}} = e^{2x}$. Both $\sin y$ and the metric coefficient are 2π-periodic. Finally, $D(\{y = k\pi\}) = \{y = (k+2)\pi\}$, proving the orbit statement. □

Remark 3.4 (No singular spine at the missing origin). The coordinate $z = 0$ is absent from $\mathbb{C}^{\times}$, but the formula $a_{\times} = Y$ and its metric extend regularly across it. Hence the origin is not a locally essential singularity in the sense of Paper I. The model is a *logarithmic-domain model*, not an essential-singularity model. Calling the missing point a singular spine would erase the local-essentiality condition.

Theorem 3.3 is the rigorous content we retain from the historical $E_{\log}$ intuition:

$$\text{countably many sheets upstairs} \quad = \quad \text{finite downstairs geometry plus deck labels.}$$

No claim is made that it reconstructs every meaning previously attached to $E_{\log}$.

3.3 Sheet labels are not automatically topological states

The label $k \in \mathbb{Z}$ in (8) depends on a choice of fundamental domain. Its parity is invariant under the deck group in this model, but an absolute integer label is not downstairs data. More generally, logarithms of moving complex roots acquire integer shifts whose individual values depend on the chosen lifts; only cycle sums survive the lift gauge. This statement is proved in Section 7 and Appendix B.

There is a second obstruction to interpreting the parallel model as a braid. If $c_1(t) < \cdots < c_k(t)$ are distinct real zero positions throughout a one-parameter family, their order cannot change without a collision. The configuration space of k unordered points on a line has contractible components; it does not carry the Artin braid group. Noncommutative braid monodromy requires motion in a surface, which in this paper comes from complex-valued fields, or else a separately declared transverse probe.

3.4 Metric completeness and provenance limits

The models above are exact solutions of the AEG equation, but completeness is not part of their claim. For example, in the parallel model the conformal factor often decays like e^{2x} as $x \to -\infty$, placing that end at finite metric distance. Completion can add boundary or singular strata and must be audited separately. This is one reason the model record must always list the domain and metric, not only the zero set.

For reference, the construction record is:

Item	Parallel model	Logarithmic model
Domain	$\mathbb{R}^2$	$\mathbb{C} \to \mathbb{C}^\times$
Assignment	$e^x h(y)$	$e^x \sin y = \operatorname{Im}(e^{x+iy})$
Metric	Equation (4)	Equation (7)
Singular set	empty	empty
Zero topology	one line per simple zero of h	countably many lines upstairs, two components downstairs
Global status	regular; completeness not asserted	regular covering model; missing origin removable
Provenance	constructed and proved here	constructed and proved here; historical name used only as motivation

Open problem 3.5 (Natural multi-zero metrics)*.* Classify multi-zero assignments when the metric is fixed by an independent AEG construction, or when completeness, curvature, or expression-history naturality is imposed. Determine whether any canonical index survives those restrictions.

4 Parameterized incidence and discriminants

4.1 Ambient families and vertical transversality

An ambient family is the correct starting object; a tube is a conclusion about a particular incidence inside it.

Definition 4.1 (Ambient AEG family)**.** An *ambient regular AEG family* over a smooth manifold B consists of a smooth fiber bundle $q : X \to B$ whose vertical tangent bundle $T^V X$ carries a smooth orientation, a smooth vertical Riemannian metric g^V, and a smooth function $A : X \to \mathbb{R}$ such that $(M_b, g_b, A_b; \mu, \lambda)$ is a regular AES for every $b \in B$. Its assignment incidence is $\mathcal{Z}(A) = A^{-1}(0)$.

Equivalently, the structure group of the surface bundle is reduced to orientation-preserving diffeomorphisms. Merely choosing an orientation on each fiber without requiring those choices to vary continuously would not orient $T^V X$ and is not sufficient here.

More generally, a *field family over an AEG background* is a rank-r real vector bundle $E \to X$ with a smooth section s. It need not be the real assignment and need not satisfy the eikonal equation. A complex-valued field is the case $r = 2$ after forgetting complex scalar multiplication.

Let $T^V X = \ker(\mathrm{d}q)$ be the vertical tangent bundle. At a zero of s, choose any connection to split TX and define the vertical derivative

$$D^V s : T^V_x X \longrightarrow E_x.$$

Its value at a zero is independent of the chosen local trivialization or connection: it is the derivative of the restricted section on the fiber.

Theorem 4.2 (Parameterized zero-section theorem)**.** *Let $q : X \to B$ be a smooth fiber bundle and $E \to X$ a rank-r vector bundle. Suppose the section $s \in \Gamma(E)$ is vertically transverse to the zero section, meaning that $D^V s$ is surjective at every $x \in \mathcal{Z}(s)$. Then:*

1. *$\mathcal{Z}(s)$ is a smooth submanifold of X of codimension r;*
2. *the restricted projection*

$$\pi = q|_{\mathcal{Z}(s)} : \mathcal{Z}(s) \longrightarrow B$$

is a smooth submersion;

3. if π is surjective and proper, then it is a locally trivial smooth fiber bundle.

The third conclusion has the standard relative form when X has boundary and the zero incidence is neat, with the boundary restriction also a submersion.

Proof. Vertical surjectivity implies surjectivity of the full derivative of s, so the zero-section transversality theorem gives the first conclusion. To prove the second, take $\xi \in T_b B$ and a lift $w \in T_x X$. Since $D^V s$ is surjective, there is $v \in T_x^V X$ with $D^V s(v) = -Ds(w)$. Then $w + v \in T_x \mathcal{Z}(s)$ and $Dq(w+v) = \xi$, so $D\pi$ is surjective. The third conclusion is Ehresmann's fibration theorem; the neat relative version follows by the corresponding boundary argument [5]. □

The rank is decisive. If the fibers of q have dimension m, then a regular zero fiber has dimension $m - r$.

Fiber dimension	Field rank	Fiberwise zero	Natural transport
2	1	curves	component and mapping-class monodromy
2	2	points	permutation and braid monodromy

For an ambient regular AEG family, equation (AEG) makes $D^V A = \mathrm{d}A_b$ nonzero everywhere, so the first two conclusions are automatic at zero. Properness is not.

4.2 The discriminant is not a single equation

For finite-dimensional polynomial root families, the algebraic discriminant detects repeated roots. A general AEG family has more ways to lose topological stability.

Definition 4.3 (Structural discriminant)**.** For a parameterized zero problem, the *structural discriminant* is the union of the following loci in the parameter space, whenever they are present:

$$\begin{aligned}
\mathcal{D}_{\mathrm{crit}} &= \{b : D^V s \text{ fails to be surjective at some zero over } b\},\\
\mathcal{D}_{\partial} &= \{b : \text{the zero incidence fails the specified neat boundary condition}\},\\
\mathcal{D}_{\mathrm{sing}} &= \{b : \text{a zero meets the declared singular stratum or the metric degenerates}\},\\
\mathcal{D}_{\mathrm{dom}} &= \{b : \text{the domain, rank, or regularity class of the field changes}\},\\
\mathcal{D}_{\infty} &= \{b : \pi \text{ is not proper over any neighborhood of } b\}.
\end{aligned}$$

If projective continuation is used, its ordinary poles and points at infinity are recorded separately rather than silently absorbed into $\mathcal{D}_{\mathrm{sing}}$.

This definition is a bookkeeping union, not a claim that every stratum is a smooth hypersurface. The geometry of its intersections and a canonical Whitney stratification remain open in general.

Corollary 4.4 (Stability off the discriminant)**.** *Let $U \subset B$ be connected. Over U, assume that $q : X \to B$ and the rank-r bundle E remain fixed smooth bundles, that s is vertically transverse to zero, and that the incidence is surjective and locally proper over U. If a boundary is present, also assume that the incidence is neat and that its boundary projection is a submersion. Then $\pi : \mathcal{Z}(s)|_U \to U$ is a smooth fiber bundle. In particular, the diffeomorphism type and number of connected components of the fiber are locally constant on U. These are precisely the operative hypotheses encoded by saying that U avoids the relevant pieces of the structural discriminant.*

Proof. For each $b \in U$, local properness supplies a neighborhood V on which the restricted incidence projection is proper. The other hypotheses make that restriction a surjective submersion, including the relative boundary condition when needed. Theorem 4.2 therefore gives a bundle chart over a possibly smaller neighborhood of b. Such charts are exactly the local-triviality assertion. □

4.3 Escape to infinity without a critical zero

The following regular AEG family shows why $\mathcal{D}_\infty$ cannot be omitted. Set

$$h_t(y) = y(ty-1), \qquad A(x,y,t) = e^x h_t(y),$$

and equip each copy of $\mathbb{R}^2$ with the smoothly varying conformal metric

$$g_t = \frac{e^{2x}(h_t(y)^2 + (2ty-1)^2)}{\mu^2 + \lambda^2 e^{2x} h_t(y)^2}(\,\mathrm{d}x^2 + \mathrm{d}y^2). \tag{9}$$

At a zero of h_t, its derivative $2ty-1$ equals -1 or 1, so it never vanishes. More strongly, h_t and $2ty-1$ never vanish simultaneously anywhere. Thus every slice is a regular AES by Theorem 2.2.

Its zero fibers are

$$Z(A_t) = \begin{cases} \{y=0\}, & t=0, \\ \{y=0\} \amalg \{y=1/t\}, & t \neq 0. \end{cases} \tag{10}$$

The total zero incidence is the disjoint union of the plane $y=0$ and the hyperbolic sheet $ty=1$. It is smooth, and its projection to t is a submersion. Nevertheless the second component escapes to spatial infinity as $t \to 0$.

Proposition 4.5 (Nonproper topology change)**.** *The family* (9) *has no critical zero and no singular metric, but the number of zero components changes at* $t=0$*. Its incidence projection is not proper over any neighborhood of* 0*.*

Proof. Only the last statement remains. For $t_j \to 0$ with $t_j \neq 0$, the points $(0, 1/t_j, t_j)$ lie in the incidence above a compact parameter interval and have no convergent subsequence. Hence the inverse image of that interval is not compact. □

Thus topology change does not require $\mathrm{d}A_t = 0$ at a finite zero. It may occur at infinity. Conversely, a smooth total incidence does not imply a tube.

4.4 Boundary control

If a surface fiber has boundary, a zero curve may be closed or may end on the boundary. For a stable relative tube one requires the incidence to meet $\partial^V X$ neatly and the restriction $\pi : \partial\mathcal{Z} \to B$ to remain a proper submersion. Without this condition a zero component can be created by tangency to the boundary even though the interior field remains regular. Such a parameter lies in $\mathcal{D}_\partial$, not in $\mathcal{D}_{\mathrm{crit}}$.

Open problem 4.6 (Stratified AEG discriminant). Under natural completeness and asymptotic hypotheses, determine when the five loci in Definition 4.3 admit a locally finite Whitney stratification and when a Thom-type isotopy lemma applies to the combined AEG data.

5 Proper zero tubes

5.1 The proper-tube theorem

Definition 1.2 reserves the word tube for a proper submersion. The resulting structure is strong and concrete.

Theorem 5.1 (Proper real-zero tube). *Let $q : X \to B$ be an ambient regular AEG family over a connected one-dimensional base, and suppose its assignment incidence $\mathcal{Z} = A^{-1}(0)$ is nonempty and $\pi = q|_{\mathcal{Z}}$ is proper and surjective.*

1. *The map $\pi : \mathcal{Z} \to B$ is a smooth fiber bundle whose fibers are smooth one-manifolds.*
2. *If the surface fibers have no boundary, every zero fiber is a finite disjoint union of circles.*
3. *If B is an interval, the bundle is globally trivial. For a product ambient family $M \times B$, its compact zero curves over each compact subinterval form a smooth isotopy and hence extend to an ambient isotopy of M. If B itself is a compact interval, this applies over all of B.*
4. *If $B = S^1$, the surface fibers are boundaryless, and the smooth vertical orientation required in Definition 4.1 is fixed, every connected component of $\mathcal{Z}$ is a torus.*

Proof. The AEG equation makes the vertical derivative $\mathrm{d}A_b$ nonzero at every point, so Theorem 4.2 proves the bundle assertion. Properness makes every fiber compact. A compact boundaryless one-manifold is a finite union of circles.

A smooth bundle over an interval is trivial. In a product ambient family, a trivialization expresses the inclusions of the compact fiber as a smooth isotopy of embeddings over each compact subinterval; the isotopy-extension theorem extends it to the ambient surface [7].

Over S^1, the total space is the mapping torus of a diffeomorphism of a finite union of circles. Each orbit of the induced component permutation produces one connected component. After traversing the orbit, the return map is a diffeomorphism of S^1, so that component is either a torus or a Klein bottle. The smooth orientation of $T^V X$ and the orientation of the base orient X; the regular level set $A^{-1}(0)$ is cooriented by $\mathrm{d}A$, hence orientable. The Klein-bottle case is excluded, and every component is a torus. This is where the global vertical-orientation hypothesis is essential: a mapping torus with orientation-reversing fiber monodromy need not be orientable. ☐

Remark 5.2 (Why vertical orientability is necessary). If the global vertical orientation is dropped, the torus conclusion is false even for a product-type formula. Let $M = \mathbb{R}_r \times S^1_\phi$ with $g = \mathrm{d}r^2 + \mathrm{d}\phi^2$, $a = \mu r$, and $\lambda = 0$, and form the mapping torus of the AEG isometry $(r, \phi) \mapsto (r, -\phi)$. Its zero incidence is the mapping torus of an orientation-reversing circle map, hence a Klein bottle. It is compact and proper over the parameter circle, but the resulting surface bundle has no smooth orientation of $T^V X$.

Remark 5.3 (What this theorem does not encode). The theorem classifies the abstract total surface under its hypotheses. The embedding $\mathcal{Z} \hookrightarrow X$ may still carry nontrivial ambient monodromy, and no preferred curve inside a torus component has been selected. In particular, a union of zero tori is not yet a knot or link.

Example 2.1 gives an immediate nonempty instance. If B is compact and $c : B \to \mathbb{R}$ is smooth, set $a_b = a_{c(b)}$. Then

$$\mathcal{Z} = \{(r, \phi, b) : r = c(b)\} \cong S^1 \times B.$$

It is compact and therefore proper over B. The example is deliberately trivial: it verifies every hypothesis while isolating the content supplied by properness.

5.2 A compact helical AEG tube

The next family is still explicit but has nontrivial component transport. Let

$$M_L = [-L, L]_x \times S^1_y, \qquad B = S^1_\theta, \qquad L > 0,$$

and choose integers $n \geq 1$ and $m \in \mathbb{Z}$. Define

$$A_{m,n}(x, y, \theta) = e^{nx} \sin(ny - m\theta) \tag{11}$$

and the vertical metric

$$g_{m,n,\theta} = \frac{n^2 e^{2nx}}{\mu^2 + \lambda^2 e^{2nx} \sin^2(ny - m\theta)} (\,\mathrm{d}x^2 + \,\mathrm{d}y^2). \tag{12}$$

Theorem 5.4 (Helical zero-tube theorem)**.** *The family (11)–(12) is an ambient regular AEG family and its zero incidence is a proper neat tube over S^1. For every θ, the zero fiber consists of $2n$ intervals*

$$y = \frac{m\theta + k\pi}{n}, \qquad k \in \mathbb{Z}/(2n). \tag{13}$$

Parameter monodromy acts on the component labels by

$$k \longmapsto k + 2m \pmod{2n}. \tag{14}$$

If $d = \gcd(n, m)$, including $d = n$ when $m = 0$, the total incidence is a disjoint union of $2d$ annuli, each containing n/d zero intervals in its monodromy orbit.

Proof. The spatial derivative is

$$\mathrm{d}^V A_{m,n} = ne^{nx} \sin(ny - m\theta)\,\mathrm{d}x + ne^{nx} \cos(ny - m\theta)\,\mathrm{d}y,$$

whose Euclidean norm squared is $n^2 e^{2nx}$. Thus (12) is precisely the conformal realization metric, proving the AEG equation. At a zero, the y-derivative is nonzero, so the incidence is smooth and meets $x = \pm L$ neatly. It is closed in the compact manifold $M_L \times S^1$, hence compact and proper.

Equation (13) follows from $ny - m\theta \in \pi\mathbb{Z}$. Increasing θ by 2π changes the label by $2m$, proving (14). Translation by $2m$ on the cyclic group of order $2n$ has $\gcd(2n, 2m) = 2d$ orbits, each of length n/d. The transport fixes the x-coordinate, so the return map after a complete label orbit is the identity on the interval. Its mapping torus is therefore an annulus, not a Möbius band. □

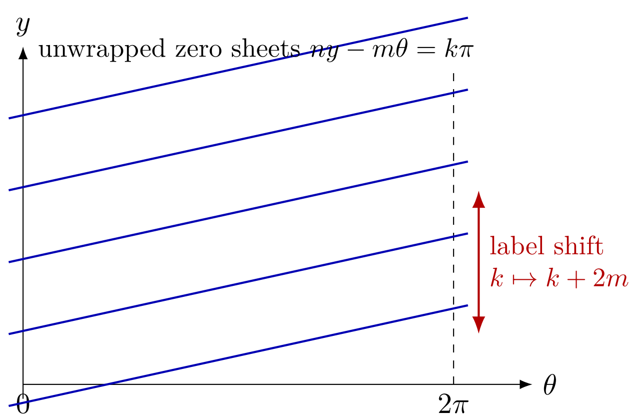

Figure 2: The helical tube on the unwrapped (θ, y)-plane. Vertical boundary identification at $\theta = 0, 2\pi$ produces the permutation (14); the drawing is schematic, while the slope is fixed exactly by $\,\mathrm{d}y/\,\mathrm{d}\theta = m/n$.

5.3 Boundary traces and logarithmic sheets

At either boundary component $x = \pm L$, the trace lies in the parameter-position torus $S^1_\theta \times S^1_y$ and satisfies

$$ny - m\theta \in \pi\mathbb{Z}.$$

It has $2d$ connected components. Each is a primitive curve with homology class

$$\frac{n}{d}[S^1_\theta] + \frac{m}{d}[S^1_y]. \tag{15}$$

If $T(p, q)$ denotes the torus link whose total homology data in these ordered coordinates are (p, q), the full trace is $T(2n, 2m)$, up to simultaneous orientation reversal. This statement depends on the declared torus coordinates; it is not an intrinsic knot type before an embedding of that torus in a three-manifold is fixed.

Unwrap the y-circle to $y \in \mathbb{R}$. Formula (13) now has labels $k \in \mathbb{Z}$. The parameter loop sends $k \mapsto k + 2m$, while the deck transformation $y \mapsto y + 2\pi$ sends $k \mapsto k + 2n$. The finite downstairs system is exactly the quotient of this countable label set by the deck action. This simultaneously realizes a proper tube and a nontrivial logarithmic shift.

Corollary 5.5 (Finite quotient of the helical lift)**.** *For the helical family, the countable lifted sheets contain no additional component permutation beyond the affine action generated by the two translations $2m$ and $2n$ on $\mathbb{Z}$. Downstairs component monodromy is their reduction modulo $2n$.*

5.4 Embedded monodromy versus braid monodromy

The permutation (14) is honest monodromy, but it is not an element of B_{2n}. Its fiber consists of intervals in a surface, not points in a disc. Intersecting the zero tube with $x = c$ produces a one-dimensional trace in $S^1_y \times S^1_\theta$, but the choice of c is additional probe data. Section 8 formalizes this distinction.

Even for compact circle fibers in Theorem 5.1, the abstract monodromy of the one-manifold fiber records only a permutation and orientation-preserving return maps of circles. The embedding in the ambient surface can carry a richer mapping class, but that is a mapping-class problem for curves, not the ordinary braid group of moving points.

6 Singular fibers and branched fillings

6.1 Real Morse events in the singular category

Let $\tau \in \mathbb{R}$ and consider the two functions

$$a^+_\tau(x, y) = x^2 + y^2 - \tau, \qquad a^-_\tau(x, y) = x^2 - y^2 - \tau. \tag{16}$$

Their common critical set in each slice is the origin. On $\mathbb{R}^2 \setminus \{0\}$ define

$$g^\pm_\tau = \frac{4(x^2 + y^2)}{\mu^2 + \lambda^2 (a^\pm_\tau)^2}(\,\mathrm{d}x^2 + \,\mathrm{d}y^2). \tag{17}$$

Proposition 6.1 (AEG Morse bifurcations)**.** *For every τ, the tuple*

$$(\mathbb{R}^2, \{0\}, g^\pm_\tau, a^\pm_\tau; \mu, \lambda)$$

is a singular AES. Its total zero incidence in $\mathbb{R}^2 \times \mathbb{R}_\tau$ is smooth, but projection to τ has a critical point at $(0,0,0)$.

For a^+_τ, the zero changes from empty to a singular point and then to a circle as τ crosses zero. For a^-_τ, the two hyperbolic branches reconnect through the four regular rays meeting at the singular origin when $\tau = 0$.

Proof. The Euclidean norm squared of $\mathrm{d}a^\pm_\tau$ is $4(x^2+y^2)$, so Corollary 2.3 gives the AEG equation and local essentiality. Define $A^\pm(x,y,\tau) = a^\pm_\tau(x,y)$. Since $\partial_\tau A^\pm = -1$, zero is a regular value of the total function; hence $(A^\pm)^{-1}(0)$ is smooth. The parameter projection restricted to it is critical precisely when the spatial derivative vanishes, which happens at the origin. The stated zero sets follow directly from $x^2+y^2=\tau$ and $x^2-y^2=\tau$. □

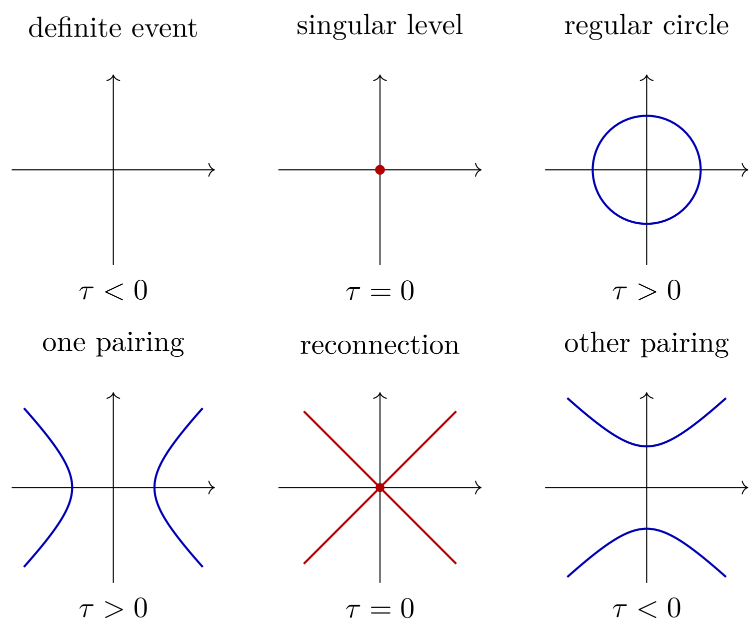

Figure 3: Real AEG-compatible Morse examples. The red point is a declared essential singularity. The pictures record zero topology only; they are not a classification of metric germs.

The proposition exhibits the key phenomenon

$$\text{smooth total zero incidence} \not\Longrightarrow \text{zero tube.}$$

At the event, it is the projection that fails to be a submersion.

6.2 A regular-to-singular fold wall

A related family makes all noncritical slices regular rather than retaining a singular point away from the zero. Put

$$A(x,y,\tau) = e^x(y^2-\tau).$$

For $\tau \neq 0$, the spatial derivative never vanishes; for $\tau = 0$, it vanishes along the entire line $y=0$. On the regular locus use

$$g_\tau = \frac{e^{2x}\left((y^2-\tau)^2+4y^2\right)}{\mu^2+\lambda^2 e^{2x}(y^2-\tau)^2}(\,\mathrm{d}x^2+\,\mathrm{d}y^2). \tag{18}$$

For $\tau < 0$ the zero fiber is empty, for $\tau > 0$ it consists of two parallel lines, and for $\tau = 0$ it is the singular line $y = 0$. The metric degenerates there and the line is locally essential. This is a verified wall-crossing family, but it is nonproper and its singular stratum is one-dimensional.

6.3 The complex branch model

Real scalar zeros and complex roots have different local models. Let $(w, \tau) \in \mathbb{C}^2$ and consider

$$F(w, \tau) = w^2 - \tau. \tag{19}$$

The incidence $F^{-1}(0)$ is a smooth complex curve parameterized by $w \mapsto (w, w^2)$. Its projection to the τ-plane is a two-sheeted branched cover with a single simple branch point at the origin.

On the boundary circle $\tau = \varepsilon e^{i\theta}$, the two roots are

$$w_{\pm}(\theta) = \pm\sqrt{\varepsilon}\, e^{i\theta/2}. \tag{20}$$

They remain distinct for every boundary parameter and exchange after one circuit. The boundary therefore carries the standard half-twist in B_2; the collision is at the center of the filling, not on the boundary braid.

Proposition 6.2 (Simple polynomial discriminant normal form)**.** *Let a holomorphic family of monic polynomials, parameterized by a complex d-manifold, meet the polynomial discriminant transversely at a parameter where exactly one root has multiplicity two and all other roots are simple. After holomorphic changes of root and parameter coordinates $\tau = (\tau_1, \ldots, \tau_d)$, its local root incidence is the disjoint union of simple graphs and one copy of*

$$w^2 = \tau_1,$$

with $\tau_2, \ldots, \tau_d$ as spectator coordinates.

Proof. Separate the simple roots by the holomorphic implicit-function theorem. The remaining quadratic factor has the form $w^2 + b(\tau)w + c(\tau)$. Translating w removes the linear term. Its discriminant is a holomorphic function of the parameter; transversality makes that function the first member τ_1 of a local coordinate system. Rescaling τ_1 gives (19), while the remaining parameter coordinates pass through unchanged. □

Restricting τ to the real axis gives two real roots for $\tau > 0$ and two imaginary roots for $\tau < 0$. Their four half-arcs meet at the branch point. This four-valent picture is a real slice of a complex branched cover, not a regular four-valent zero of a real assignment.

6.4 Smooth and holomorphic fillings

Section 7 realizes every braid by a family that is smooth in the loop parameter and holomorphic in the spatial coordinate w. Extending such a family over a parameter disc while remaining holomorphic in both variables is a stronger demand. Algebraic-function boundaries are constrained by quasipositivity [8]. Thus the statements

every braid has an arithmetic-holomorphic spatial realization,

every braid bounds a holomorphic AEG root filling

are not equivalent; the first is proved below, while the second is false without restrictions.

Open problem 6.3 (AEG-natural singular fillings)*.* Determine which branch signs and global braided surfaces arise when the parameter dependence is constrained by expression history or by a natural connection, rather than chosen as an arbitrary smooth or holomorphic polynomial family.

7 Complex roots, monodromy, and braids

7.1 Square-free polynomials as a finite-root interface

Let $\mathcal{P}_n \cong \mathbb{C}^n$ be the space of monic degree-n complex polynomials and let $\Delta_n \subset \mathcal{P}_n$ be its algebraic discriminant. The square-free locus is

$$\mathcal{P}_n^{\mathrm{sf}} = \mathcal{P}_n \setminus \Delta_n.$$

Vieta's map identifies it with the unordered configuration space

$$\mathcal{P}_n^{\mathrm{sf}} \cong \mathrm{UConf}_n(\mathbb{C}) = \mathrm{Conf}_n(\mathbb{C})/S_n. \tag{21}$$

Its fundamental group is the Artin braid group B_n [2, 6, 3].

Theorem 7.1 (Finite-root incidence and braid monodromy)**.** *Let B be a connected smooth manifold and let*

$$F_b(w) = w^n + c_1(b)w^{n-1} + \cdots + c_n(b) \tag{22}$$

be a smooth family of monic polynomials with nonvanishing discriminant. Then

$$\mathcal{R}_F = \{(w, b) \in \mathbb{C} \times B : F_b(w) = 0\} \tag{23}$$

is a smooth real codimension-two submanifold whose projection to B is a proper n-sheeted covering. The coefficient map $c : B \to \mathcal{P}_n^{\mathrm{sf}}$ induces $c_* : \pi_1(B, b_0) \to \pi_1(\mathcal{P}_n^{\mathrm{sf}}, c(b_0))$. After choosing the usual path to a standard base configuration and the resulting identification of the target with B_n, this is braid monodromy

$$c_* : \pi_1(B, b_0) \longrightarrow B_n. \tag{24}$$

Changing that identification conjugates the representation. Its composition with $B_n \to S_n$ is the ordinary permutation monodromy of the covering $\mathcal{R}_F \to B$.

Proof. At a root of a square-free polynomial, $\partial F_b/\partial w \neq 0$. The implicit-function theorem therefore writes the root incidence locally as n disjoint graphs over B, so it is an n-sheeted covering. A finite covering of locally compact Hausdorff spaces is proper. The map to unordered root configurations is exactly the coefficient map under (21); functoriality of the fundamental group gives (24). Forgetting paths and retaining only their endpoint labels is the standard quotient $B_n \to S_n$. $\square$

For $B = S^1$, the incidence is a closed geometric braid in $\mathbb{C} \times S^1$. A basepoint and an ordering above it give an element of B_n; changing the basepoint or ordering conjugates that element. This is strictly more information than the endpoint permutation.

Warning 7.2 (Rank-one assignments do not yield B_n)*.* The field F in Theorem 7.1 is complex-valued and has real rank two. The theorem does not turn a real AEG assignment into a polynomial, nor does it promote a moving family of zero curves to a finite-root braid. The two constructions share the parameterized zero-section theorem but have different codimensions.

7.2 Arithmetic-holomorphic braid realization

On the complete basic hyperbolic AES with $\mu, \lambda > 0$, Paper II constructs a global coordinate $w = X + iy \in \mathbb{H}$ in which arithmetic holomorphicity is ordinary holomorphicity. This supplies a precise AEG background for Theorem 7.1.

Theorem 7.3 (AEG realization of every braid)**.** *For every $\beta \in B_n$ and every complete basic hyperbolic AES with $\mu, \lambda > 0$, there is a smooth loop of arithmetic-holomorphic fields F_θ such that:*

1. *every F_θ is a monic degree-n polynomial in the arithmetic holomorphic coordinate w;*
2. *all roots lie in a fixed disc $D \Subset \mathbb{H}$ and remain simple;*
3. *the root incidence realizes the geometric braid β.*

Consequently, every oriented link is the closure of the root incidence of such an arithmetic-holomorphic family.

Proof. Represent β by a smooth loop $\{z_1(\theta), \ldots, z_n(\theta)\}$ in $\mathrm{UConf}_n(\mathbb{C})$. On the parameter interval $[0, 2\pi]$, choose a smooth ordered lift $(z_1(\theta), \ldots, z_n(\theta))$; its two endpoints may differ by the permutation underlying β. An affine translation and positive rescaling place the compact unordered image in a disc $D \Subset \mathbb{H}$ without changing its braid class. Set

$$F_\theta(w) = \prod_{j=1}^{n} (w - z_j(\theta)). \tag{25}$$

Because the endpoint lists differ only by a permutation, the coefficients are smooth and periodic in θ. The roots are simple, and each slice is holomorphic in w. Paper II identifies it as an arithmetic-holomorphic field. Its root incidence is the chosen loop of configurations. The last statement follows from Alexander's closed-braid theorem [1]. □

The theorem is a universality result, not an invariant. Because every braid can be realized, the predicate "admits an AEG root-tube realization" cannot distinguish links. New information would have to be supplied by a natural history-derived decoration or by a restricted class of allowed families.

7.3 Logarithmic lifting of root tubes

Suppose all roots avoid the origin. Pull the root incidence back along $\exp : \mathbb{C} \to \mathbb{C}^\times$:

$$\widetilde{\mathcal{R}}_F = \{(b, \omega) : F_b(e^\omega) = 0\}. \tag{26}$$

Theorem 7.4 (Logarithmic root transport and gauge)**.** *The map $\widetilde{\mathcal{R}}_F \to \mathcal{R}_F$ is the canonical principal $2\pi i\mathbb{Z}$-cover obtained by pullback of the exponential cover. Its specified deck-translation action is $\omega \mapsto \omega + 2\pi i k$. Along a parameter loop, choose a labeling of the initial roots and logarithmic lifts. If root continuation has permutation σ, then there are integers k_i such that*

$$\omega_i(1) = \omega_{\sigma(i)}(0) + 2\pi i k_i. \tag{27}$$

Changing the initial lifts by $\omega_i(0) \mapsto \omega_i(0) + 2\pi i m_i$ changes the integers by

$$k_i \longmapsto k_i + m_i - m_{\sigma(i)}. \tag{28}$$

For every cycle of σ, the sum of the k_i on that cycle is gauge invariant. Relabeling the roots permutes these cycle sums, so their multiset, rather than an ordered list, is label-independent. The total sum $\sum_i k_i$ is gauge and label invariant and equals the winding of the product of the roots around zero.

Proof. Equation (26) is the pullback of the exponential cover, so the first claim is immediate. Unique path lifting gives (27). If the initial lift of the ith path is shifted by $2\pi i m_i$, its entire lift is shifted by that amount. Comparing the new endpoint with the new lift assigned to the root $\sigma(i)$ gives (28). The m_i terms telescope around each permutation cycle. Finally, the product of all roots is, up to sign, the constant coefficient of F_b; summing logarithmic increments computes its winding. □

Thus a vector (k_i) is not an invariant before a lift gauge is fixed. Cycle sums are invariant but abelian. The full noncommutative information remains in the configuration path, more precisely in the annular braid group when roots are constrained to $\mathbb{C}^\times$.

7.4 Boundary braid and discriminant filling

Theorem 7.1 applies only in the complement of Δ_n. A filling disc may meet the discriminant. At a transverse simple intersection, Proposition 6.2 gives $w^2 = \tau$, and its boundary loop contributes a conjugate of a standard positive or negative half-twist according to orientation. The collision is therefore a two-cell event in the filling, while the boundary braid stays collision-free.

This separation resolves an otherwise persistent ambiguity:

Object	Allowed local event	Output
regular real assignment family	no critical zero	curve tube
singular real family	Morse critical zero	birth or reconnection
square-free boundary root family	no repeated root	braid monodromy
root filling through Δ_n	simple double root	branch half-twist

Open problem 7.5 (Natural braid monodromy)*.* Find a functorial procedure that assigns a finite complex-root field to an expression history or to a real zero tube, and prove that the resulting braid is independent of auxiliary coordinates and probes. The arbitrary polynomial realization of Theorem 7.3 does not solve this problem.

8 Threading, affine torsion, and the knot frontier

8.1 Threads are additional data

The real zero tube has curve fibers. To obtain moving points one must select them.

Definition 8.1 (Finite zero thread)**.** Let $\pi : \mathcal{Z} \to B$ be an embedded real zero tube over a connected one-manifold. A *finite zero thread* is a properly embedded one-dimensional submanifold $K \subset \mathcal{Z}$, with $\partial K = K \cap \pi^{-1}(\partial B)$ when the base has boundary, such that $\pi|_K : K \to B$ is a finite-sheeted covering. For $B = S^1$, this forces K to be a compact one-manifold without boundary. The pair $(\mathcal{Z}, K)$ is a *threaded zero tube.* A thread is *intrinsic* only if its construction from the stated AEG data is functorial under the declared notion of AEG isomorphism.

In the helical family, every slice $x = c$ gives

$$K_c = \mathcal{Z}(A_{m,n}) \cap \{x = c\},$$

a degree-$2n$ finite thread. Its homology and component structure are computed in (15). The choice of c is a probe choice; if one uses $c = \pm L$, the boundary is part of the domain data but the choice of boundary component remains. Selecting only one of the $2d$ components is another choice.

A rank-two root incidence such as $\mathcal{R}_F$ is already one-dimensional, so it plays the role of its own finite thread. If its moving points lie in a disc, it is an ordinary geometric braid. If they lie in an annulus or general surface, the relevant surface braid group and equivalence relation must be named.

8.2 A scalar Markov obstruction

The first novelty filter is independent of AEG.

Proposition 8.2 (Stateless additive collapse)**.** *Let A be an abelian group and $n \geq 2$. Every homomorphism $\Theta_n : B_n \to A$ factors through the exponent sum (writhe) $w : B_n \to \mathbb{Z}$:*

$$\Theta_n(\beta) = w(\beta)c_n$$

for a fixed $c_n \in A$. If the family Θ_n is compatible with the standard inclusions $B_n \hookrightarrow B_{n+1}$ and is unchanged under both Markov stabilizations $\beta \mapsto \beta\sigma_n^{\pm 1}$, then every Θ_n is zero.

Proof. In an abelian target, the braid relation $\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}$ implies $\Theta_n(\sigma_i) = \Theta_n(\sigma_{i+1})$. All standard generators therefore have the same image c_n, and the homomorphism is exponent sum times c_n. Inclusion compatibility identifies c_n with c_{n+1}. Positive stabilization then gives $c_{n+1} = 0$; negative stabilization gives the same conclusion. Hence the family vanishes. □

Allowing a normalization depending on strand number does not evade both signs. If $I_n(\beta) = c\,w(\beta) + dn$ in an additive coefficient group, positive and negative stabilization require $c + d = 0$ and $-c+d = 0$. Away from two-torsion this forces $c = d = 0$, and in all cases it leaves no noncommutative information. A useful AEG invariant must retain state, use a trace-like normalization, or belong to a more structured category.

8.3 Affine conjugation and the Alexander layer

Let R be a commutative ring. Write an affine transformation as (p, u), with $p \in R^\times$ and $u \in R$, acting by $x \mapsto px + u$. Thus all multipliers p, q, r below are units and all translation coordinates u, v, w lie in R. We use

$$(p, u)(q, v) = (pq, u + pv), \qquad (p, u)^{-1} = (p^{-1}, -p^{-1}u). \tag{29}$$

Right conjugation $x \triangleright y = y^{-1}xy$ is

$$(p, u) \triangleright (q, v) = \Big(p, q^{-1}(u + (p-1)v)\Big). \tag{30}$$

On the fixed-multiplier branch $X_t = \{(t, u) : u \in R\}$ over a commutative ring in which t is a unit, this becomes

$$u \triangleright v = t^{-1}u + (1 - t^{-1})v. \tag{31}$$

It is exactly an Alexander quandle, not a structure beyond the Alexander layer.

The affine torsion expression

$$\kappa\big((p, u), (q, v)\big) = (1 - q)u + (p - 1)v \tag{32}$$

restricts to

$$\kappa_t(u, v) = (t - 1)(v - u). \tag{33}$$

Proposition 8.3 (Ordinary exactness)**.** *With the ordinary quandle coboundary convention $df(x, y) = f(x \triangleright y) - f(x)$, one has*

$$\kappa_t = d(tf) \qquad \textit{for } f(u) = u.$$

Thus the fixed-multiplier affine torsion represents the zero ordinary quandle cohomology class.

Proof. Equation (31) gives

$$df(u, v) = (t^{-1} - 1)u + (1 - t^{-1})v = \frac{t-1}{t}(v - u).$$

Multiplication by t gives (33). □

8.4 Twisting, resonance, and a nonzero class

We now use the twisted convention of Carter–Elhamdadi–Saito [4]. Let the coefficient action be a scalar T_{coef}. A normalized twisted 2-cocycle ϕ satisfies

$$\begin{aligned} T_{\mathrm{coef}}\phi(x,y) + \phi(x \triangleright y, z) = T_{\mathrm{coef}}\phi(x,z) &+ (1-T_{\mathrm{coef}})\phi(y,z) \\ &+ \phi(x \triangleright z, y \triangleright z), \end{aligned} \tag{34}$$

and $\phi(x,x) = 0$. For coboundaries we choose the sign

$$\delta_T^+ f(x,y) = T_{\mathrm{coef}} f(x) + (1-T_{\mathrm{coef}}) f(y) - f(x \triangleright y). \tag{35}$$

The chain-level convention in the cited paper differs from (35) by a global minus sign in one displayed definition, while its extension and state-sum calculations use the sign above. The coboundary subgroup is the same under either choice.

Put $\alpha = t^{-1}$. Direct substitution in (34) shows that κ_t is a twisted cocycle for every coefficient action. If $\alpha - T_{\mathrm{coef}}$ is a unit in the coefficient ring, it is exact: for

$$f(u) = \frac{t-1}{\alpha - T_{\mathrm{coef}}} u$$

one has $\delta_T^+ f = \kappa_t$. Over a field, only the resonance $T_{\mathrm{coef}} = \alpha$ survives this linear exactness test; over a general ring, a nonzero nonunit difference may also obstruct this particular linear primitive.

Theorem 8.4 (Finite-field resonant affine class)**.** *Let q be a prime power, let $t \in \mathbb{F}_q^\times \setminus \{1\}$, and put $\alpha = t^{-1}$. Give $X_t = \mathbb{F}_q$ the Alexander operation $x \triangleright y = \alpha x + (1-\alpha) y$ and give the coefficient module $N = \mathbb{F}_q$ the resonant action $T_{\mathrm{coef}} = \alpha$. Then*

$$\kappa_t(x,y) = (t-1)(y-x) \tag{36}$$

is a normalized twisted 2-cocycle representing a nonzero class in $H^2_{TQ}(X_t; N)$.

Proof. Normalization is immediate, and substituting the Alexander operation into (34) makes both sides equal

$$(t-1)\big[z - (\alpha + T_{\mathrm{coef}})x + (\alpha + T_{\mathrm{coef}} - 1)y\big].$$

Suppose $\kappa_t = \delta_T^+ f$. Subtracting the constant $f(0)$ does not change the coboundary, so take $g(0) = 0$. Setting $y = 0$ gives

$$g(\alpha x) - \alpha g(x) = (t-1)x. \tag{37}$$

Let $r = \mathrm{ord}(\alpha)$. Iterating (37) yields

$$g(\alpha^r x) = \alpha^r g(x) + r(t-1)\alpha^{r-1} x.$$

But $\alpha^r = 1$ and r divides $q-1$, so the characteristic of $\mathbb{F}_q$ does not divide r. Taking $x \neq 0$ contradicts $t \neq 1$. Hence no one-cochain has coboundary κ_t. $\square$

This closes the algebraic resonance question, but it does not yet produce a stronger classical knot invariant. In fact, a second calculation gives a sharp collapse theorem.

We now fix the precise planar state-sum convention. Give each oriented arc the normal for which the ordered pair (tangent, normal) agrees with the plane orientation, and give the unbounded region Alexander number zero. At a crossing τ, let $L(\tau)$ be the Alexander number of the source region from which both arc normals point. Let y_τ be the over-arc color, and let its normal point from an under-arc colored x_τ to one colored $x_\tau \triangleright y_\tau$. Define $\epsilon(\tau) = 1$ when the ordered pair (over-arc normal, under-arc normal) agrees with the plane orientation and $\epsilon(\tau) = -1$ otherwise. In additive coefficient notation the twisted Boltzmann weight is

$$B_T(\tau, C) = T_{\mathrm{coef}}^{-L(\tau)}(\epsilon(\tau)\kappa_t(x_\tau, y_\tau)) \in N, \tag{38}$$

and the state sum is

$$\Phi_{\kappa_t}(D) = \sum_{C \in \mathrm{Col}_{X_t}(D)} \left[\sum_\tau B_T(\tau, C)\right] \in \mathbb{Z}[N]. \tag{39}$$

Here $[a]$ denotes the basis element of the integral group ring indexed by $a \in N$. This is the additive form of the standard twisted convention.

Theorem 8.5 (Planar state-sum collapse)**.** *Under the hypotheses of Theorem 8.4, the standard twisted quandle cocycle state sum* (39) *is a Reidemeister invariant of oriented classical links, but for every planar link diagram D it is*

$$\Phi_{\kappa_t}(D) = |\mathrm{Col}_{X_t}(D)|[0]. \tag{40}$$

Thus it provides no enhancement of the Alexander quandle coloring count.

Proof. The Reidemeister invariance follows from the standard twisted-cocycle theorem [4]. To identify the value, define the two-dimensional $\mathbb{F}_q$-vector space $E = N \oplus X_t$ with Laurent action

$$T_E(a, x) = (\alpha a - (t-1)x, \alpha x). \tag{41}$$

This is invertible and makes $0 \to N \to E \to X_t \to 0$ a short exact sequence of Alexander modules. For the section $s(x) = (0, x)$,

$$T_E s(x) + (1 - T_E)s(y) - s(x \triangleright y) = (\kappa_t(x, y), 0). \tag{42}$$

Hence κ_t is the obstruction cocycle of an Alexander-module extension. The planar obstruction-cocycle cancellation theorem of Carter–Elhamdadi–Saito then makes every coloring contribute the identity weight. With convention (39), the state sum is therefore exactly $|\operatorname{Col}_{X_t}(D)|[0]$. $\square$

Theorems 8.4 and 8.5 illustrate why cohomological nontriviality and knot-theoretic detection must be tested separately. A nonzero class can have a trivial planar enhancement because its weights cancel through an extension. On diagrams in nonplanar surfaces, the same extension argument need not cancel; that is a different, surface-link question and is not promoted here to an ordinary S^3 invariant.

8.5 Variable multipliers and the Reidemeister-III anomaly

For the full affine quandle, the torsion (32) is generally not an ordinary 2-cocycle. With $x = (p, u)$, $y = (q, v)$, and $z = (r, w)$, define its type-III defect by

$$\mathfrak{A}_\kappa(x, y, z) = \kappa(x, y) + \kappa(x \triangleright y, z) - \kappa(x, z) - \kappa(x \triangleright z, y \triangleright z).$$

A direct calculation gives

$$\mathfrak{A}_\kappa(x, y, z) = \frac{(q-r)(r-1)}{qr}\,\kappa(x, y). \tag{43}$$

It vanishes on a fixed-multiplier branch, as it must. On a one-component knot, conjugate meridians also have equal multipliers, so arbitrary multiplier variation along arcs is not legitimate coloring data. Nonzero use of (43) would require additional region state, a multi-component setting, a dynamical quandle, or another declared structure.

Warning 8.6 (Anomaly is not associator). A failure of the ordinary type-III cocycle equation is not automatically a higher associator or a surface-holonomy invariant. Such an interpretation requires separate pentagon/hexagon or equivalent coherence and filling-independence proofs.

8.6 A historical free-word calculation

One useful repository computation can be stated without overclaiming it. Use the historical figure-eight knot-group presentation

$$G_{4_1} = \langle a, b \mid abbbaBAAB = 1 \rangle, \qquad A = a^{-1}, \quad B = b^{-1}.$$

Under the convention that $a : x \mapsto tx$, $b : x \mapsto x + 1$, uppercase letters denote inverses, and the displayed transformations compose in the fixed word order, the free-group word

$$abbbaBAAB$$

has unit multiplier and translation coordinate

$$-\Delta_{4_1}(t) = -(t^2 - 3t + 1). \tag{44}$$

This is a verified affine word calculation. The displayed generator assignment descends from the free group to G_{4_1} only on the parameter locus $\Delta_{4_1}(t) = 0$. At generic t it is not a knot invariant; it is the failure of the chosen generator assignment to satisfy the relator.

8.7 The remaining invariant gate

Theorem 7.3 supplies every braid, and Definition 8.1 supplies explicit threads in selected real tubes. Neither result selects a choice-free numerical or algebraic decoration. For a family $I_n : B_n \to \mathcal{A}$ to descend to oriented links via Markov's theorem, it must at minimum satisfy

$$\begin{aligned} I_n(\gamma\beta\gamma^{-1}) &= I_n(\beta), \\ I_{n+1}(\iota_n\beta\,\sigma_n) &= I_n(\beta), \\ I_{n+1}(\iota_n\beta\,\sigma_n^{-1}) &= I_n(\beta). \end{aligned} \tag{45}$$

after any declared normalization [3]. An AEG construction must additionally prove independence of logarithmic lift, probe, thread, spine, axis, basepoint, and expression presentation whenever those are auxiliary choices.

Open problem 8.7 (AEG Markov descent). Construct a nonabelian or stateful decoration natural under AEG isomorphisms, prove the braid relations and all choice-independence statements, and find a Markov normalization satisfying (45).

Open problem 8.8 (Separation beyond Alexander and Burau). After Problem 8.7 is solved, exhibit two oriented links with the same declared Alexander/Burau baseline but different AEG invariant. A stronger braid representation without closure descent does not meet this criterion.

8.8 Conclusion

The tube programme now has a rigorous geometric core. Regular real assignments produce curve tubes under properness; their compact circle-parameter total spaces are tori. The helical family makes component transport, deck lifting, and selected threads explicit. Singular AEG metrics realize the two elementary real Morse events, while complex root fillings isolate the branch half-twist. Paper II's function theory realizes every braid by arithmetic-holomorphic roots.

The same analysis locates the frontier. Flexible metrics make existence abundant, and universal braid realization makes mere representability nondiscriminating. Stateless scalars collapse to writhe. Fixed affine multipliers reduce to an Alexander quandle; their ordinary torsion is exact. Resonant twisting gives a genuinely nonzero cohomology class, but its planar classical-link state sum still collapses to the coloring count. The remaining original problem is therefore not to draw another tube, but to derive a natural decoration from arithmetic history and prove that it survives every quotient required to become a knot invariant.

A Metric, regularity, and properness calculations

A.1 Conformal scaling identity

If $g = \rho h$ on a surface, then $g^{-1} = \rho^{-1}h^{-1}$. Therefore, for any smooth real function a,

$$|\nabla a|_g^2 = |\,\mathrm{d}a|_{g^{-1}}^2 = \rho^{-1}|\,\mathrm{d}a|_h^2. \tag{46}$$

Taking $\rho = |\,\mathrm{d}a|_h^2/(\mu^2 + \lambda^2 a^2)$ proves Theorem 2.2. The construction is valid exactly where $\mathrm{d}a \neq 0$; at a critical point the displayed coefficient vanishes and is not a Riemannian metric.

For $a_h = e^x h(y)$,

$$\partial_x a_h = e^x h, \qquad \partial_y a_h = e^x h',$$

which gives Equation (4). For the helical assignment, writing $\xi = ny - m\theta$ gives

$$\partial_x A = ne^{nx}\sin\xi, \qquad \partial_y A = ne^{nx}\cos\xi,$$

so $|\,\mathrm{d}^V A|^2 = n^2 e^{2nx}$ independently of ξ. This cancellation is the reason the helical metric has the particularly simple numerator in (12).

A.2 Vertical transversality in coordinates

Locally trivialize $q : X \to B$ and $E \to X$ near a zero so that $X \cong U \times V$ and $E \cong (U \times V) \times \mathbb{R}^r$, with $U \subset \mathbb{R}^m$ a fiber chart and $V \subset \mathbb{R}^k$ a base chart. The section becomes a map $s(u, b) \in \mathbb{R}^r$. Vertical transversality says that the $r \times m$ matrix $D_u s$ has rank r. The full Jacobian $(D_u s, D_b s)$ then has rank r, proving that $s^{-1}(0)$ has dimension $m + k - r$.

To see the projection submersion explicitly, solve

$$D_u s\,\dot{u} = -D_b s\,\dot{b}$$

for $\dot{u}$ using surjectivity of $D_u s$. Then $(\dot{u}, \dot{b})$ is tangent to the incidence and projects to the arbitrary vector $\dot{b}$. This is the local linear-algebra content of Theorem 4.2.

A.3 Properness checks

For the helical model, $M_L \times S^1$ is compact and the zero incidence is the inverse image of the closed set $\{0\}$ under a continuous function. It is therefore compact, and every map from it to the Hausdorff circle is proper. At $x = \pm L$, the vertical normal to the boundary is ∂_x, while $\partial_y A \neq 0$ on the zero set. Hence the zero incidence is neat and its boundary projects submersively.

For the escape family of Proposition 4.5, fix any $\varepsilon > 0$ and set $t_j = \varepsilon/(j+1)$. The sequence

$$(x_j, y_j, t_j) = (0, t_j^{-1}, t_j)$$

lies in the zero incidence above $[0, \varepsilon]$ and leaves every compact subset of $\mathbb{R}^2 \times [0, \varepsilon]$. This is a direct certificate that the incidence projection is not proper.

A.4 Audit records for the explicit models

The following records make all data and limitations explicit.

Cylindrical product model. Domain $\mathbb{R} \times S^1$; metric $\mathrm{d}r^2 + \mathrm{d}\phi^2$; assignment $(\mu/\lambda)\sinh(\lambda(r-c))$ for $\lambda \neq 0$ and $\mu(r-c)$ for $\lambda = 0$; singular set empty; equation verified by differentiation; zero topology one circle; compact parameter base gives a proper product incidence; status: proved.

Parallel multi-zero model. Domain $\mathbb{R}^2$; metric (4); assignment $e^x h(y)$; singular set empty when all zeros of h are simple; equation verified by conformal realization; zero topology one line for each zero of h; no parameter required; completeness not asserted; status: proved, newly constructed in this paper.

Logarithmic model. Downstairs domain $\mathbb{C}^\times$, upstairs domain $\mathbb{C}$; metrics and assignments in (6) and (7); singular set empty; equation verified by pullback; two zero components downstairs and countably many upstairs; deck action $k \mapsto k+2$; missing origin removable; status: proved covering model, not an essential singular AES.

Helical model. Domain $[-L, L] \times S^1$; parameter $\theta \in S^1$; metric (12); assignment (11); singular set empty; equation verified above; each zero fiber has $2n$ intervals; total incidence has $2\gcd(n, m)$ annuli and is proper with neat boundary; status: proved.

Morse models. Domain $\mathbb{R}^2$, singular set $\{0\}$, parameter $\tau \in \mathbb{R}$; metric (17); assignments (16); equation verified on the punctured plane; definite birth/death and indefinite reconnection; the total zero incidence is smooth but its parameter projection is critical; status: proved examples, not universal AEG normal forms.

Complex branch model. Background is the basic hyperbolic AES from Paper II with $\mu, \lambda > 0$; field $F_\tau(w) = w^2 - \tau$ is complex-valued and arithmetic holomorphic in w; the field is not the real assignment; root incidence is smooth, projection has one simple branch point; boundary gives a half-twist; status: proved local root model.

A.5 Completeness warning

Conformal realization does not preserve completeness. If a conformal coefficient ρ decays rapidly along an end, an h-infinite path may have finite $g = \rho h$ length. In the parallel examples, the factor e^{2x} commonly makes $x \to -\infty$ a finite-distance end. Any theorem that needs complete fibers must therefore add completeness as a hypothesis and recheck each model; it cannot infer it from the eikonal identity.

B Configuration spaces, logarithmic lifts, and branch signs

B.1 Configuration and braid conventions

The ordered configuration space of n points in a surface S is

$$\mathrm{Conf}_n(S) = \{(z_1, \dots, z_n) \in S^n : z_i \neq z_j \text{ for } i \neq j\},$$

and $\mathrm{UConf}_n(S) = \mathrm{Conf}_n(S)/S_n$. For the plane, $\pi_1(\mathrm{UConf}_n(\mathbb{C})) = B_n$. Our generators σ_i exchange the ith and $(i+1)$st points by a positive half-twist and satisfy

$$\begin{aligned} \sigma_i\sigma_j &= \sigma_j\sigma_i && (|i-j| \geq 2), \\ \sigma_i\sigma_{i+1}\sigma_i &= \sigma_{i+1}\sigma_i\sigma_{i+1}. \end{aligned}$$

The quotient $B_n \to S_n$ remembers only endpoint permutation, and the abelianization $B_n \to \mathbb{Z}$ sends every σ_i to 1.

The identification (21) sends an unordered set of roots to the elementary symmetric coefficients of its monic polynomial. It is a homeomorphism, with smooth complex-manifold structure on the square-free locus. The classical configuration-space fibrations supply one standard route to the fundamental-group computation [6].

B.2 Properness of a finite root covering

Let $p : Y \to B$ be a finite-sheeted covering with B locally compact Hausdorff. For a compact set $K \subset B$, cover K by finitely many relatively compact evenly covered neighborhoods. The closure of each sufficiently small sheet is homeomorphic to a compact closure downstairs. A finite union contains $p^{-1}(K)$ as a closed subset, so $p^{-1}(K)$ is compact. Thus p is proper. Applied to the root incidence, this proof uses square-freeness locally and the fixed finite degree globally. Monicity prevents a root from disappearing to infinity while the coefficient remains at a fixed point of B.

B.3 Gauge calculation for logarithmic roots

Let $z_i(s)$ be the continuation path beginning at the ith labeled root. If $z_i(1) = z_{\sigma(i)}(0)$ and ω_i is its logarithmic lift, then

$$\omega_i(1) - \omega_{\sigma(i)}(0) \in 2\pi i\mathbb{Z}.$$

This defines k_i. Shift initial lifts by $m_j \in \mathbb{Z}$. Unique path lifting gives

$$\omega_i'(s) = \omega_i(s) + 2\pi i m_i,$$

and hence

$$\begin{aligned} \omega_i'(1) &= \omega_{\sigma(i)}(0) + 2\pi i(k_i + m_i) \\ &= \omega_{\sigma(i)}'(0) + 2\pi i(k_i + m_i - m_{\sigma(i)}). \end{aligned}$$

This proves (28). If $i_1 \mapsto i_2 \mapsto \cdots \mapsto i_\ell \mapsto i_1$ is a permutation cycle, then

$$\sum_{j=1}^{\ell}(m_{i_j} - m_{\sigma(i_j)}) = 0.$$

The cycle sum of deck shifts is therefore lift-gauge invariant, whereas each summand is not. Relabeling the roots permutes the cycles, so the unordered multiset of cycle sums is canonical.

The total sum has an algebraic expression. If the roots are $z_i(b)$, then

$$\prod_i z_i(b) = (-1)^n c_n(b).$$

The winding of the constant coefficient around zero equals the sum of the root windings and hence $\sum_i k_i$.

B.4 Helical trace on the parameter-position torus

On $S^1_\theta \times S^1_y$, the equation $ny - m\theta \in \pi\mathbb{Z}$ is the inverse image of the two values 0 and π under the group homomorphism

$$(\theta, y) \longmapsto ny - m\theta \pmod{2\pi}.$$

The kernel of the homomorphism with coefficient vector $(-m, n)$ has $d = \gcd(n, m)$ components. Each of the two inverse images is a translate of that kernel, giving $2d$ components in total. A primitive parameterization of any component is

$$s \longmapsto \left(\theta_0 + \frac{n}{d}s,\, y_0 + \frac{m}{d}s\right), \qquad s \in \mathbb{R}/(2\pi\mathbb{Z}),$$

which proves (15). With the ordered torus-link convention declared in Section 5, the union is $T(2n, 2m)$.

B.5 The sign of the simple half-twist

For $w^2 = \tau$ on the positively oriented boundary $\tau = \varepsilon e^{i\theta}$, choose at $\theta = 0$ the ordering $(-\sqrt{\varepsilon}, +\sqrt{\varepsilon})$. As θ increases, both points turn counterclockwise through angle π and exchange. Under the convention that a counterclockwise exchange in the root plane is positive, the braid is σ_1. Reversing the boundary orientation gives σ_1^{-1}. Changing the local coordinate w by an orientation-preserving complex affine map does not change this sign.

This local sign does not determine a global filling. If a parameter disc meets the discriminant in several transverse points, the boundary braid is a product of conjugates of signed half-twists, with order determined by a choice of paths from a basepoint. Holomorphic dependence restricts the possible signs and factorizations; the quasipositive boundary theorem is the relevant classical constraint [8].

B.6 Closure equivalence

Alexander's theorem says that every oriented link in S^3 has a closed-braid representative. Markov's theorem says that two braids have isotopic oriented closures precisely after a sequence generated by braid conjugation and positive or negative stabilization, allowing changes of strand number. Accordingly, a quantity on B_n is not a classical link invariant until the three identities in (45), with its chosen normalization, are proved. If the braid axis is retained, one obtains an annular invariant under a smaller equivalence relation; it must not be relabeled an ordinary S^3 invariant.

C Affine and twisted-quandle calculations

C.1 Conjugation in the affine group

From (29),

$$\begin{aligned}(q,v)^{-1}(p,u)(q,v) &= (q^{-1},-q^{-1}v)(p,u)(q,v)\\ &= \left(p, q^{-1}(u-v)+q^{-1}pv\right)\\ &= \left(p, q^{-1}\big(u+(p-1)v\big)\right),\end{aligned}$$

which proves (30). The multiplier of the left input is preserved by conjugation. Consequently each fixed-multiplier set is a subquandle, and taking $p=q=t$ gives (31).

The torsion expression can be recognized as the difference between the translation parts of the two affine products. More precisely, define the translation-coordinate projection by $\mathrm{trans}(p,u)=u$. Then

$$\mathrm{trans}\big((p,u)(q,v)\big)-\mathrm{trans}\big((q,v)(p,u)\big)=(1-q)u+(p-1)v.$$

This verifies (32) and its fixed-branch restriction.

C.2 Twisted cocycle and primitive

Write $c=t-1$, $\alpha=t^{-1}$, and let the coefficient action be multiplication by β. For $\phi(x,y)=c(y-x)$, the left-hand side of (34), divided by c, is

$$\beta(y-x)+z-(\alpha x+(1-\alpha)y)=z-(\alpha+\beta)x+(\alpha+\beta-1)y.$$

The right-hand side is

$$\begin{aligned}&\beta(z-x)+(1-\beta)(z-y)\\ &\quad+(\alpha y+(1-\alpha)z)-(\alpha x+(1-\alpha)z)\\ &=z-(\alpha+\beta)x+(\alpha+\beta-1)y.\end{aligned}$$

Thus ϕ is a twisted cocycle for every β.

For $f(x)=Cx$, convention (35) gives

$$\delta_T^+ f(x,y)=C(\alpha-\beta)(y-x).$$

If $\alpha-\beta$ is invertible, choosing $C=c/(\alpha-\beta)$ proves nonresonant exactness. At $\beta=\alpha$, this calculation vanishes for every linear f, but Theorem 8.4 is stronger: it excludes every set-theoretic one-cochain over the finite field.

For completeness, the induction used there follows from $g(\alpha x)=\alpha g(x)+cx$:

$$\begin{aligned}g(\alpha^{j+1}x)&=\alpha g(\alpha^j x)+c\alpha^j x\\ &=\alpha^{j+1}g(x)+(j+1)c\alpha^j x.\end{aligned}$$

C.3 The extension calculation

Relative to the ordered basis of $E=N\oplus X_t$, the operator in (41) is

$$[T_E]=\begin{pmatrix}\alpha & -(t-1)\\ 0 & \alpha\end{pmatrix},\qquad \det[T_E]=\alpha^2\neq 0.$$

Thus it defines an Alexander module. With $s(x) = (0, x)$,

$$T_E s(x) = (-(t-1)x, \alpha x),$$
$$(1 - T_E)s(y) = ((t-1)y, (1-\alpha)y).$$

Their sum is $(\kappa_t(x, y), x \triangleright y)$, which proves (42). Equivalently, the Alexander operation on E is

$$(a, x) * (b, y) = \big(\alpha a + (1-\alpha)b + \kappa_t(x, y), x \triangleright y\big).$$

This is the explicit extension behind Theorem 8.5.

C.4 Two coloring-count checks

The state sum uses crossing weights modified by the Alexander numbering of the source region. The extension theorem makes the total weight zero for every planar coloring, so only the number of colorings remains. Two small cases check the result.

For the positive two-strand braid, use the coloring map

$$B(a, b) = (b, a \triangleright b).$$

Writing $d = b - a$, iteration gives $d_{j+1} = -\alpha d_j$.

The closure of B^2 is the positive Hopf link. Its fixed-point equation forces $(1-\alpha)d = 0$, so $d = 0$ because $t \neq 1$. Hence

$$\Phi_{\kappa_t}(H) = q[0]. \tag{47}$$

The closure of B^3 is the positive trefoil. Its fixed-point equation is $(\alpha^2 - \alpha + 1)d = 0$, giving

$$\Phi_{\kappa_t}(3_1) = \begin{cases} q^2[0], & \alpha^2 - \alpha + 1 = 0, \\ q[0], & \alpha^2 - \alpha + 1 \neq 0. \end{cases} \tag{48}$$

For $q = 3$ and $t = \alpha = 2$, the values are $3[0]$ and $9[0]$. Their difference is entirely the Alexander-quandle coloring count, not a cocycle enhancement.

C.5 Variable-multiplier defect

Let $K = \kappa(x, y)$. From (30),

$$\kappa(x \triangleright y, z) - \kappa(x, z) = \frac{1-r}{q} K,$$
$$\kappa(x \triangleright z, y \triangleright z) = \frac{1}{r} K.$$

Therefore

$$\mathfrak{A}_\kappa(x, y, z) = \left(1 + \frac{1-r}{q} - \frac{1}{r}\right) K$$
$$= \frac{(q-r)(r-1)}{qr} K,$$

which proves (43). The translation coordinate w of z cancels identically; this does not provide the higher coherence absent from the construction.

C.6 The figure-eight word

With $a = (t, 0)$, $b = (1, 1)$, $A = a^{-1}$, and $B = b^{-1}$, multiply the letters of $abbbaBAAB$ from left to right using (29):

Prefix	Multiplier	Translation
a	t	0
ab	t	t
abb	t	$2t$
$abbb$	t	$3t$
$abbba$	t^2	$3t$
$abbbaB$	t^2	$3t - t^2$
$abbbaBA$	t	$3t - t^2$
$abbbaBAA$	1	$3t - t^2$
$abbbaBAAB$	1	$3t - t^2 - 1$

The final translation is $-(t^2 - 3t + 1)$, proving (44) under the declared convention.

References

[1] James W. Alexander. A lemma on systems of knotted curves. *Proceedings of the National Academy of Sciences of the United States of America*, 9(3):93–95, 1923.

[2] Emil Artin. Theorie der zöpfe. *Abhandlungen aus dem Mathematischen Seminar der Universität Hamburg*, 4:47–72, 1925.

[3] Joan S. Birman. *Braids, Links, and Mapping Class Groups*, volume 82 of *Annals of Mathematics Studies*. Princeton University Press, Princeton, NJ, 1974.

[4] J. Scott Carter, Mohamed Elhamdadi, and Masahico Saito. Twisted quandle homology theory and cocycle knot invariants. *Algebraic & Geometric Topology*, 2:95–135, 2002.

[5] Charles Ehresmann. Les connexions infinitésimales dans un espace fibré différentiable. In *Séminaire Bourbaki, 1948/49–1950/51*, volume 1, pages 153–168. Société Mathématique de France, 1952. Exposé no. 24.

[6] Edward Fadell and Lee Neuwirth. Configuration spaces. *Mathematica Scandinavica*, 10:111–118, 1962.

[7] Morris W. Hirsch. *Differential Topology*, volume 33 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1976.

[8] Lee Rudolph. Algebraic functions and closed braids. *Topology*, 22(2):191–202, 1983.

[9] Mingli Yuan. Arithmetic expression geometry i: Foundations—sequential histories, affine flow, torsion, and contact geometry. Mathematical-review manuscript, 2026.

[10] Mingli Yuan. Arithmetic expression geometry ii: Hyperbolic real function theory— horizontal operators, boundary problems, and arithmetic holomorphicity. Mathematical-review manuscript, 2026.